Beyond Climatology: Thermal Tolerance Landscapes Forecast the Biological Impacts of Marine Heatwaves

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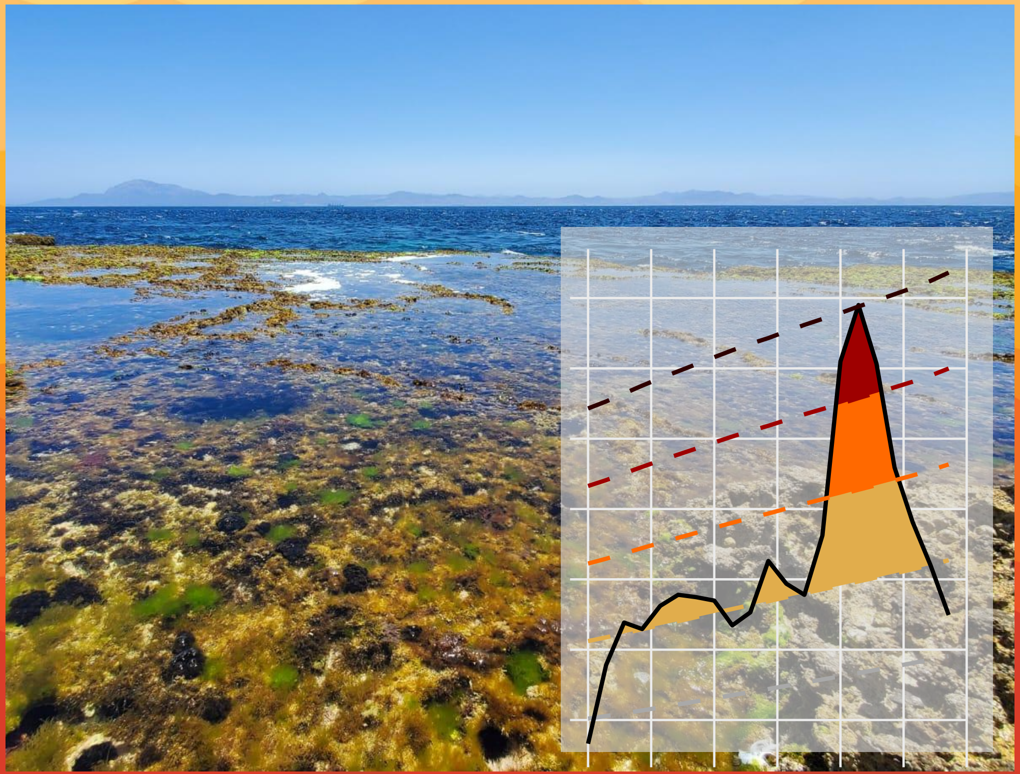
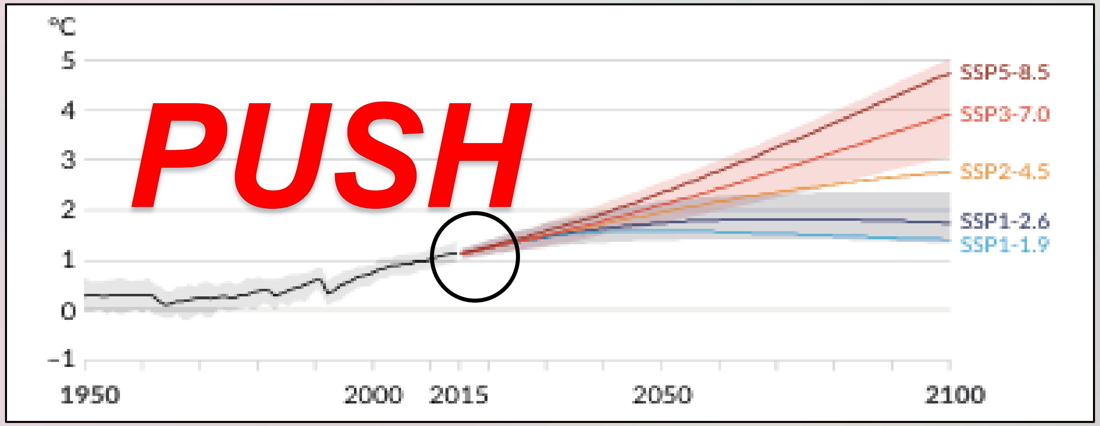


B51B-05 – Ecological Forecasting in Managed and Natural Systems

American Geophysical Union 2025 Meeting
19 December 2025

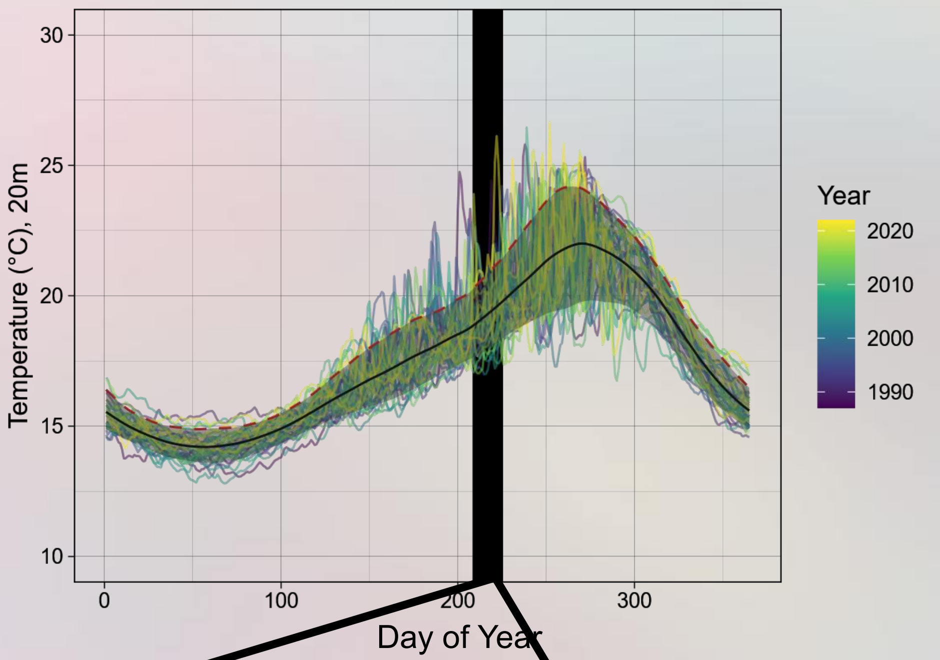
PREDICTING POPULATIONS UNDER A VARIABLE FUTURE

IPCC RCP Global mean temperature projections



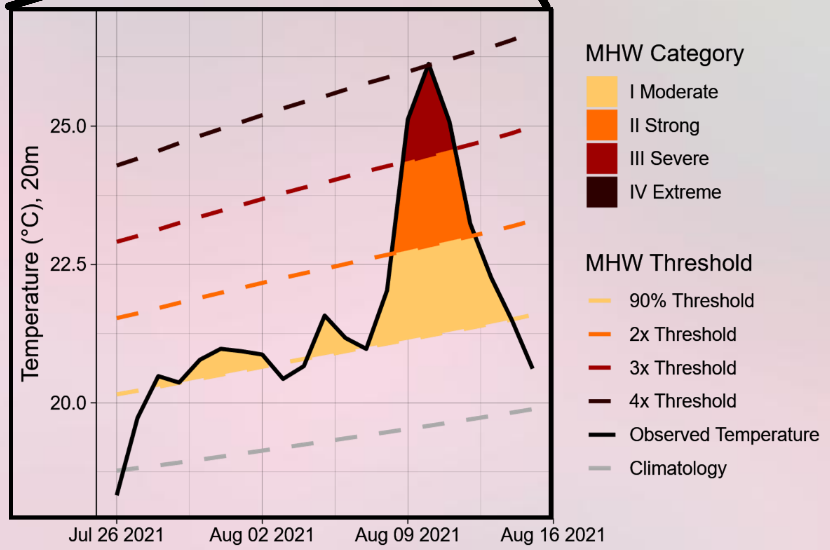
- The rise of extreme heatwaves over the push of climate change challenges predictions of the future.
- We see extreme temperature events happening on much quicker time scales – pulses occurring minutes to months (Oliver et al. 2019).
- This is the timescale at which many organisms operate (Sasaki et al. 2025).

TRANSLATING HEATWAVES: FROM STATISTICS TO BIOLOGY



<u>Marine Heatwaves (MHWs)</u>

Temperature magnitude exceeds 90th percentile of 30-year historical time series – describes what is extreme for a given location (Hobday et al. 2016, 2018).



But what is extreme for organisms?

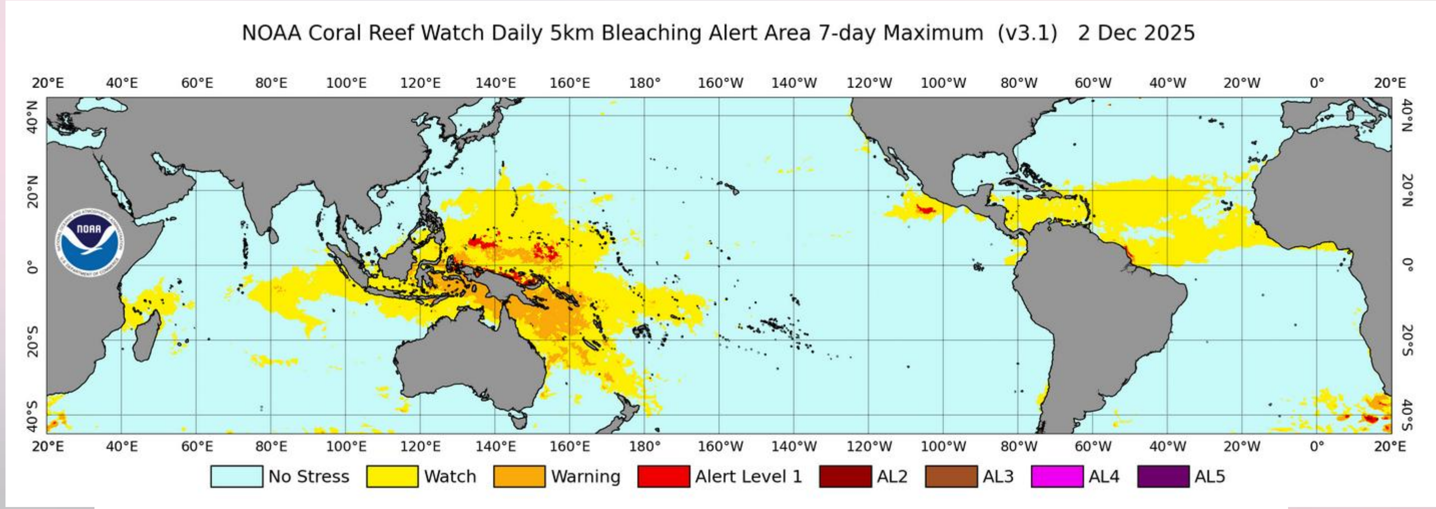
Subtidal Gorgonians, Mediterranean Sea, 2003 & 2018

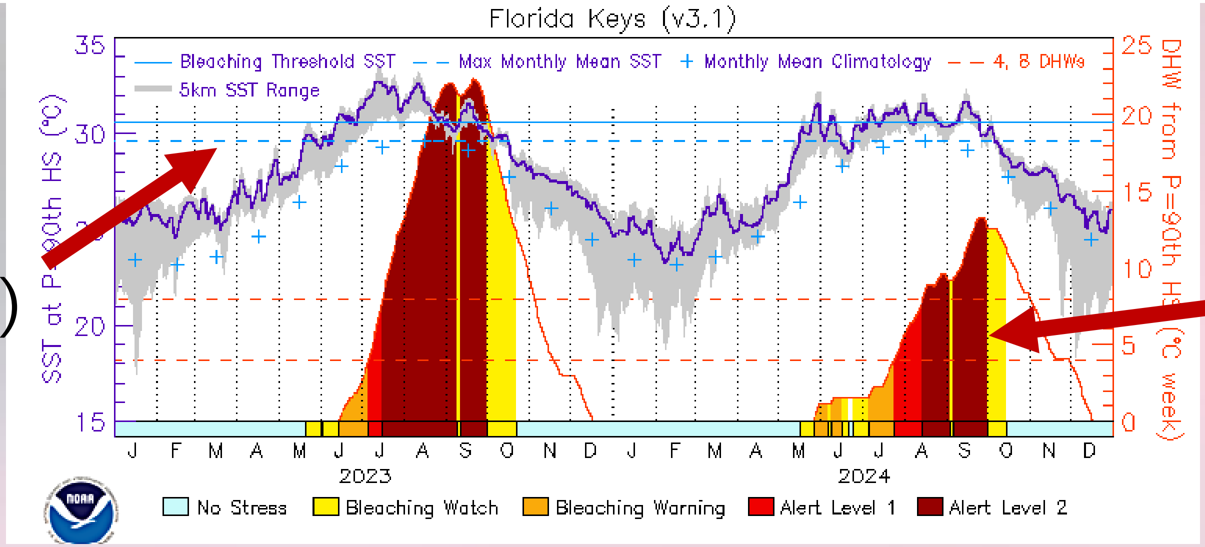
Garrabou et al. 2022

**Forecasting Goals: 1) Predict when conditions cause mass mortality events,
and 2) what the extent of damage resulting from these events**

Smith KE, et al. 2023
Annu. Rev. Mar. Sci. 15:119–45

CURRENT FORECASTS OF BIOLOGICAL HEAT STRESS ARE STILL BASED ON STATIC THRESHOLDS AND CLIMATOLOGICAL ANALYSIS, NOT BIOLOGY!





Static Threshold (Max Monthly SST)

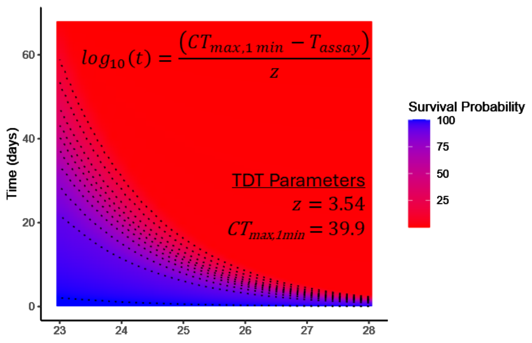
Linear stress accumulation

Skirving et al. 2020

A BIOLOGICAL APPROACH: THERMAL TOLERANCE LANDSCAPES

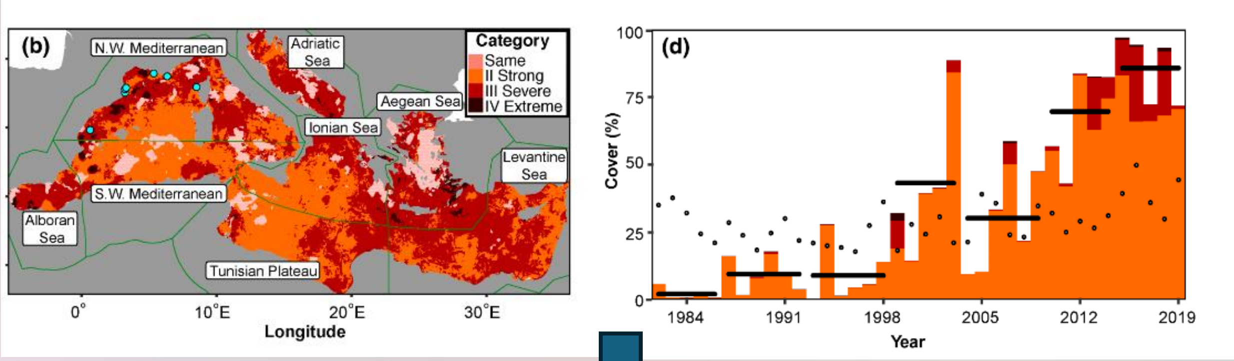
Villeneuve and White 2024, Journal of Animal Ecology

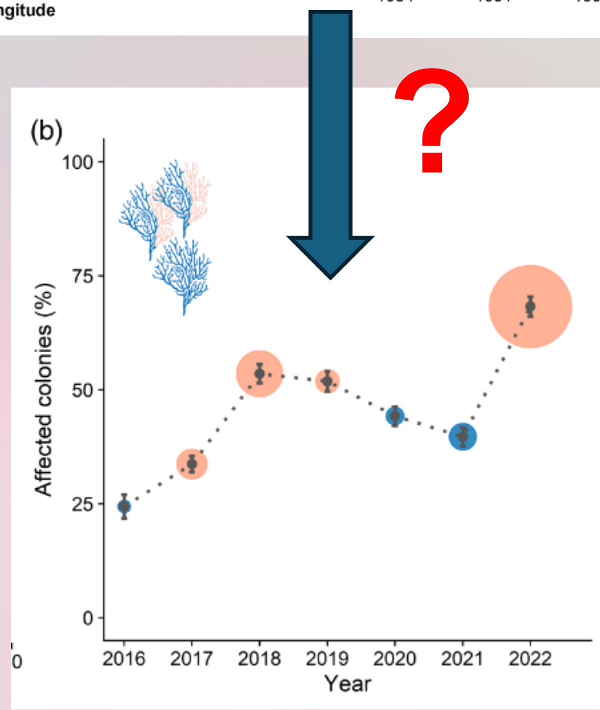
- It matters how *long* you spend at stressful temperatures – temperature as a dose (Rezende et al. 2014).
- Calculate **injury accumulation** using dynamic survival models (Rezende et al. 2020, Jørgensen et al. 2021, Ørsted et al. 2024).
  - Inputs: TTL and field body temperature
- For a given heatwave, we can calculate end survival and assess severity.

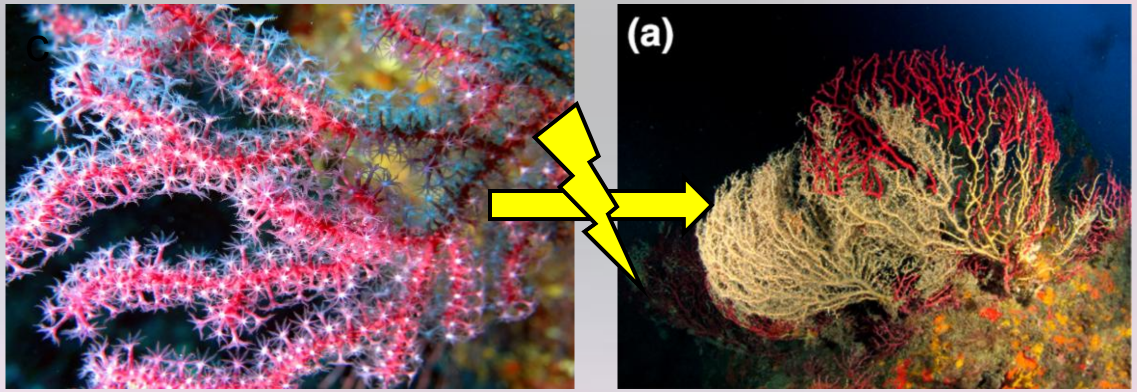


MAPPING BASIN-WIDE MORTALITY OF THE GORGONIAN *PARAMURICEA CLAVATA* DURING MHW EVENTS

- Intense marine heatwaves in 1999, 2003, and 2022 - extensive mass mortality.
- What types of MHWs are the most common? Which are the most “risky” based on occurrence probability and mortality risk?



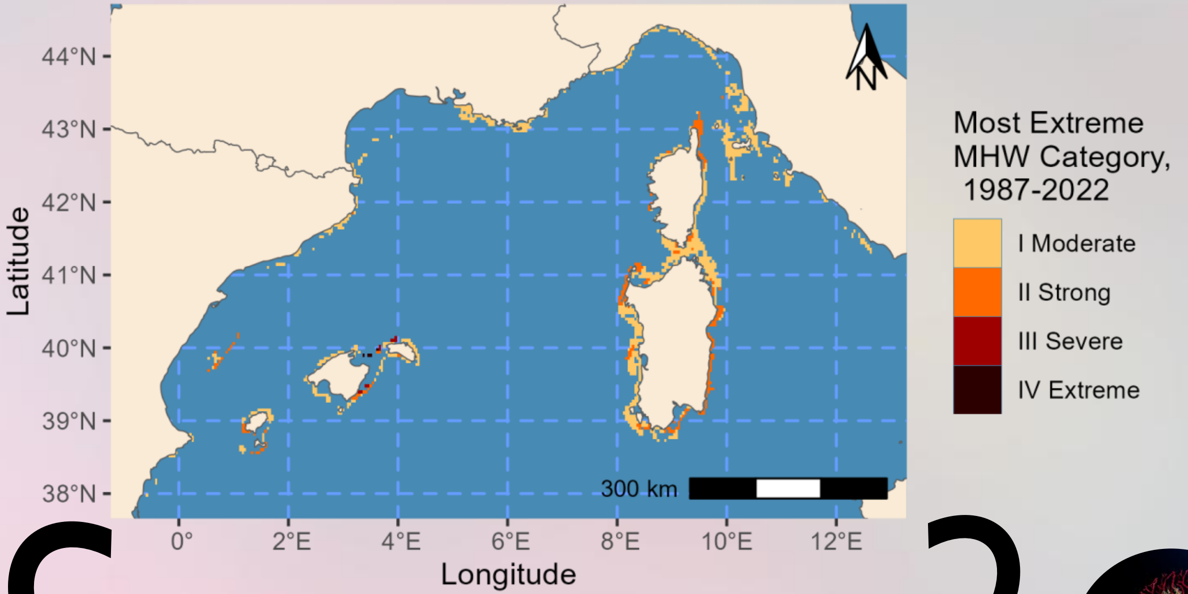


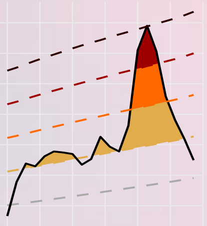


Garrabou et al. 2022

MAPPING BASIN-WIDE MORTALITY OF THE GORGONIAN *PARAMURICEA CLAVATA*

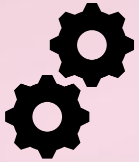
- **Temperature:** Copernicus 4.2km Reanalysis and Forecast ocean temperature (20m depth), clipped to species range.
- **Thermal Tolerance Landscape:** created using data in the literature (Pairaud 2014, Gómez-Gras 2021).
- **Approach:** Identify <u>climatological heatwaves</u> in each range grid cell, then calculate and categorize <u>mortality</u> <u>severity</u>.



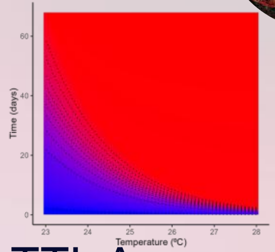


Statistical MHW Analysis

Heat Injury Algorithm

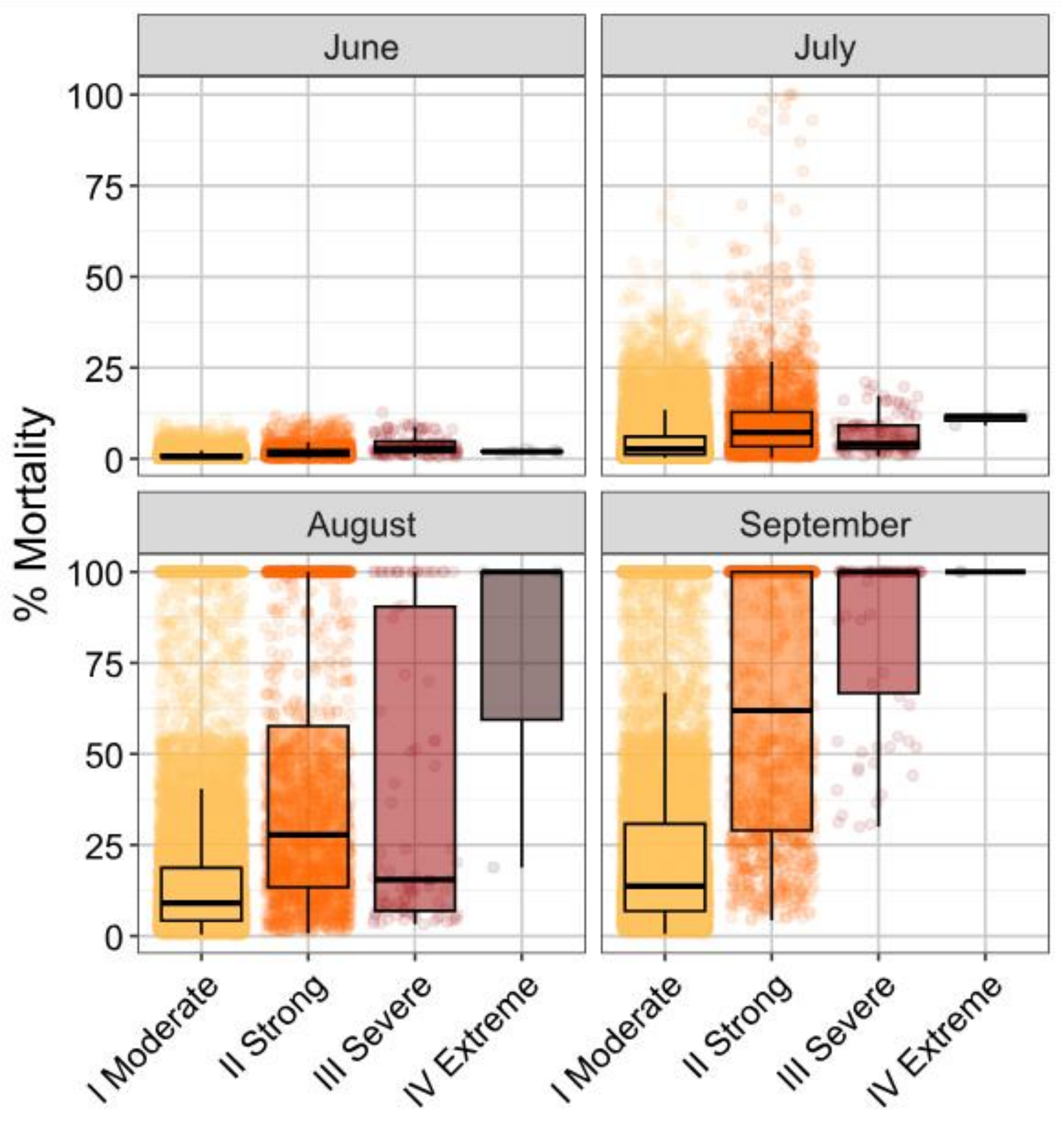


Survival



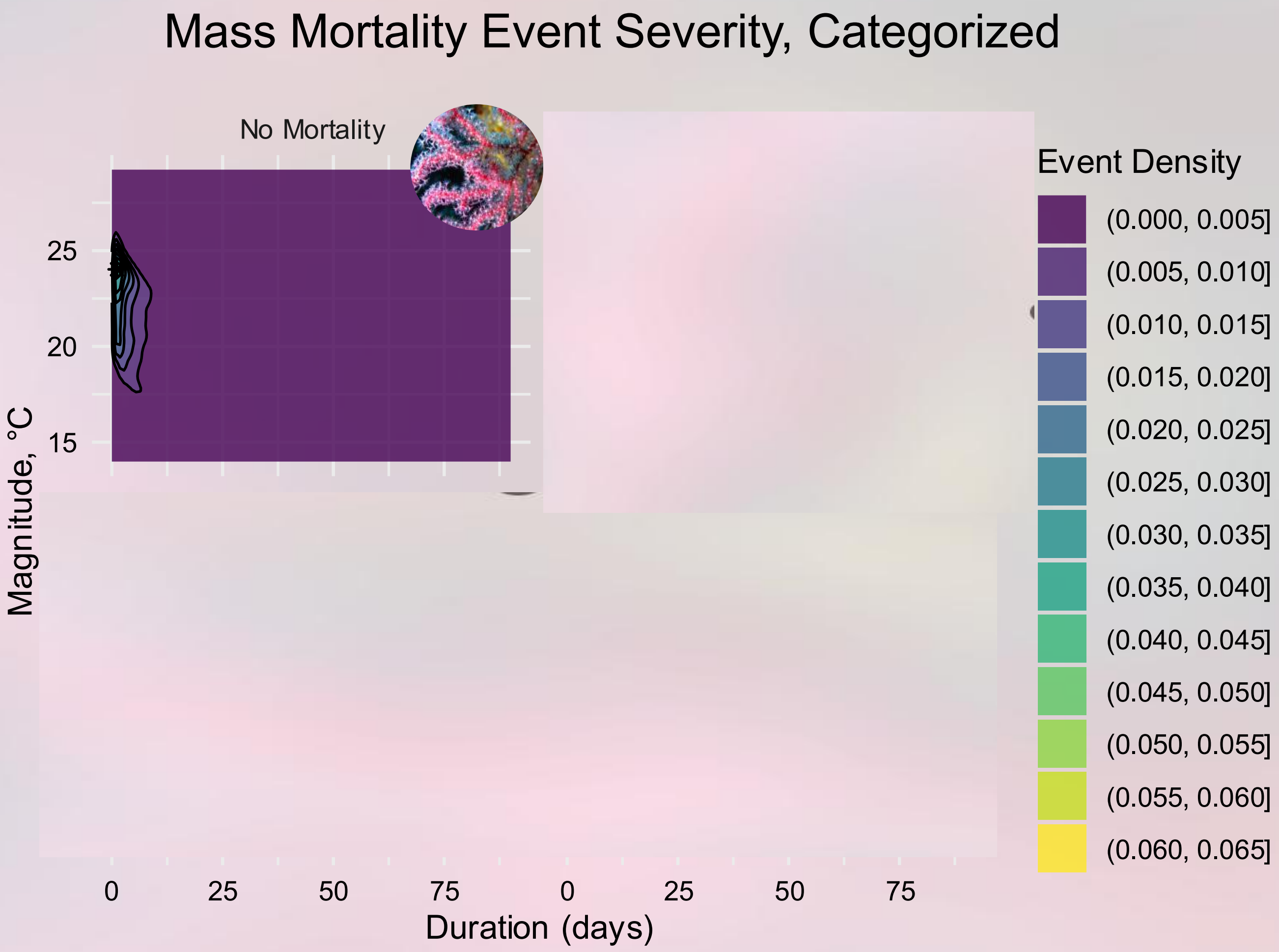
TTL Approach

MHW SEVERITY IS NOT USUALLY ALIGNED WITH MORTALITY SEVERITY



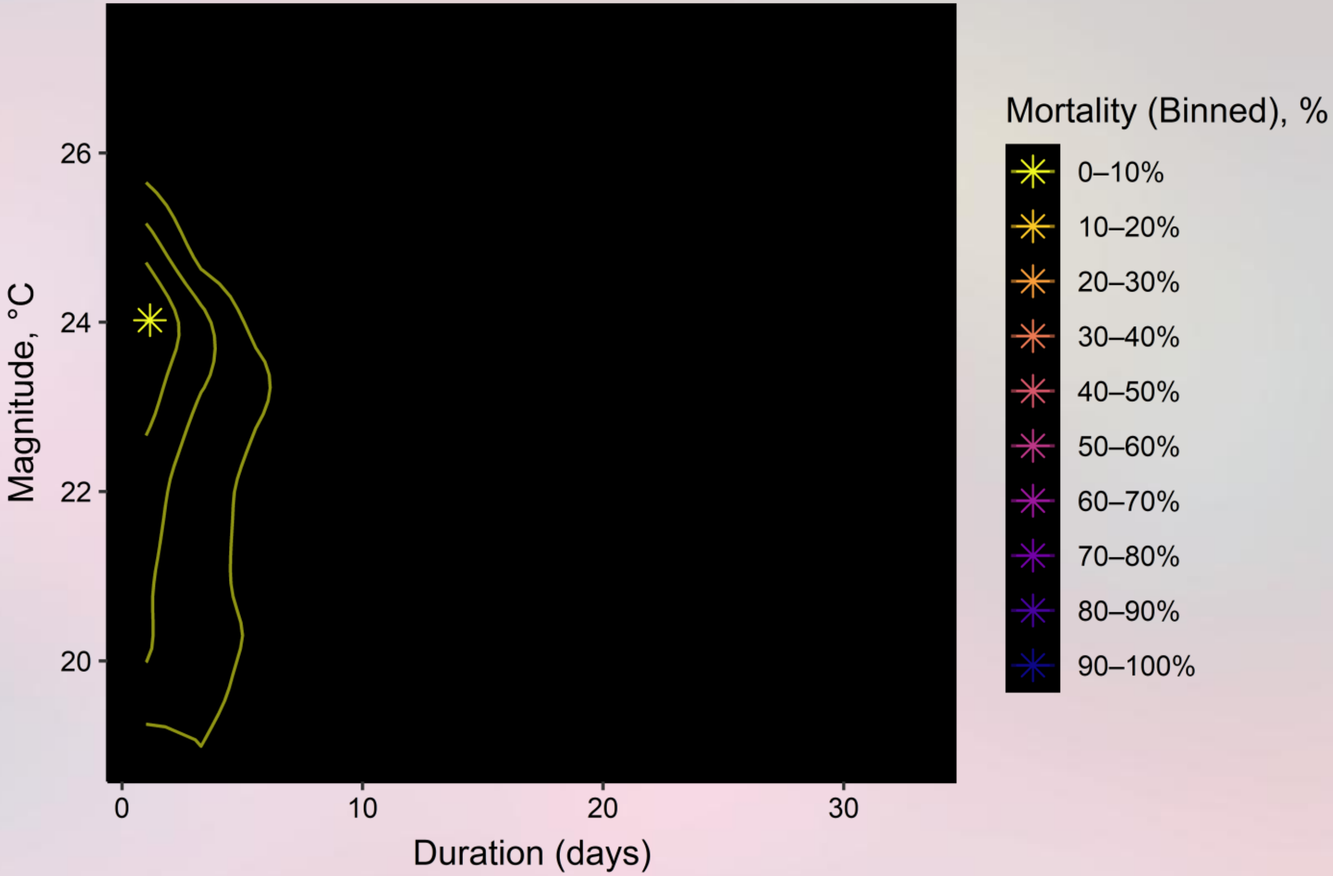
DIFFERENT MHW EVENTS CAN HAVE SIMILAR IMPACTS

- Survival isoclines exist in modeled survival data – corals with similar mortality can occur over a range of MHW types.



DIFFERENT MHW EVENTS CAN HAVE SIMILAR IMPACTS

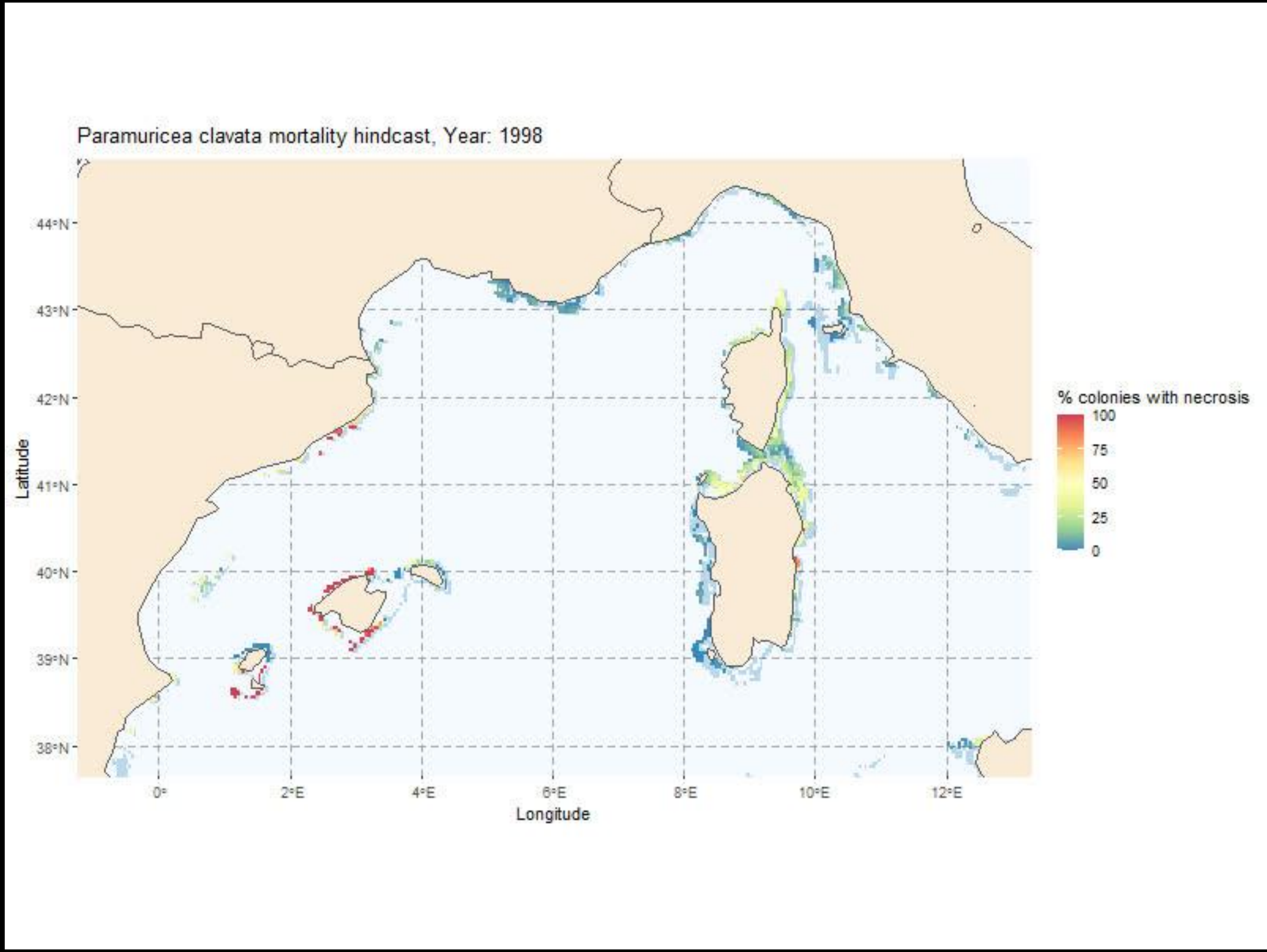
- Survival isoclines exist in modeled survival data – corals with similar mortality can occur over a range of MHW types.
- Mortality centroid shifts with MHWs



Towards Operational Forecasts of Extreme Event Biological Impacts

Forecasts at the same temporal and spatial scale as our oceanographic data

Remote monitoring of hard-to-survey areas



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Sea Grant NEW HAMPSHIRE

University of New Hampshire NH Agricultural Experiment Station

University of New Hampshire School of Marine Science and Ocean Engineering

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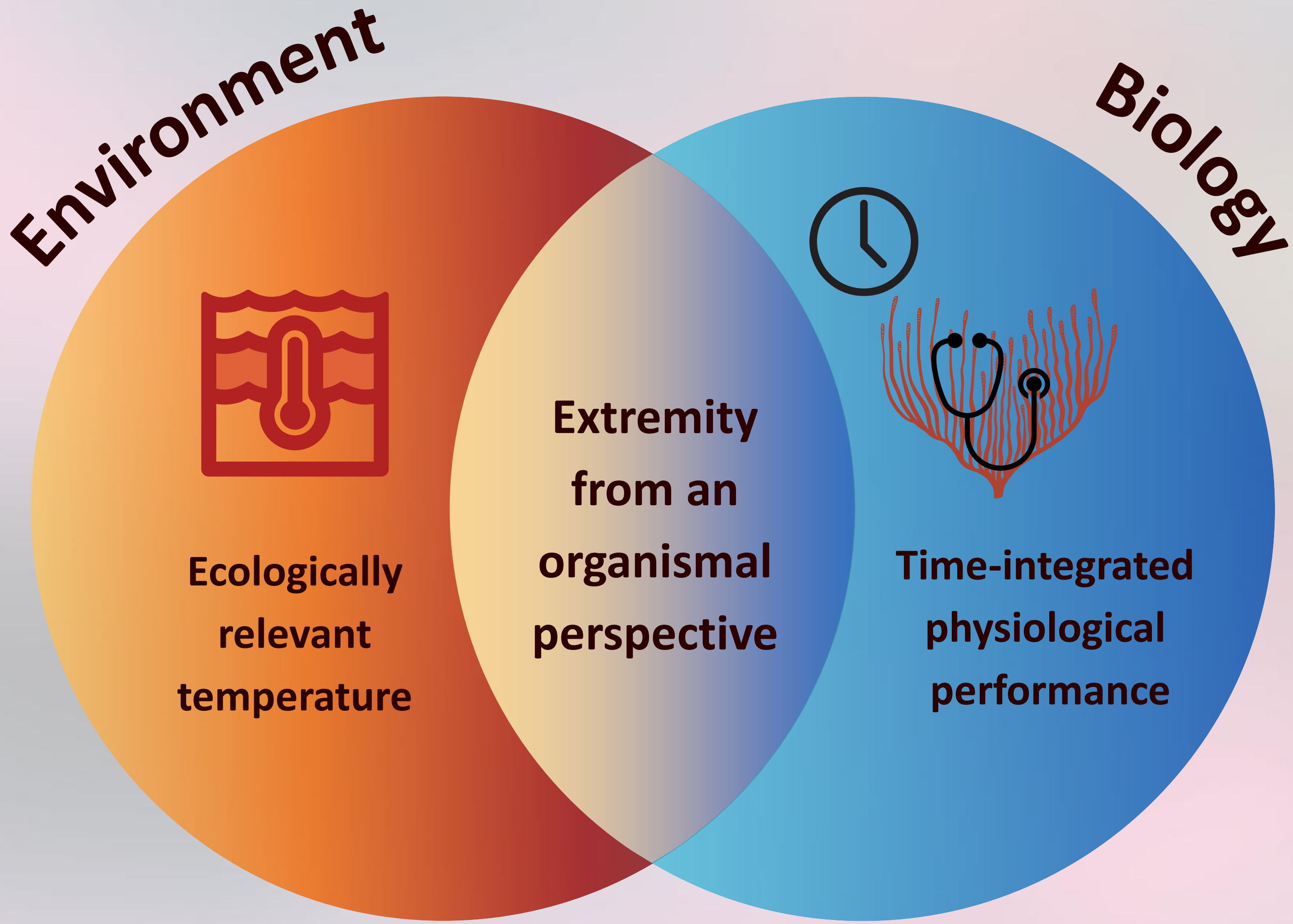
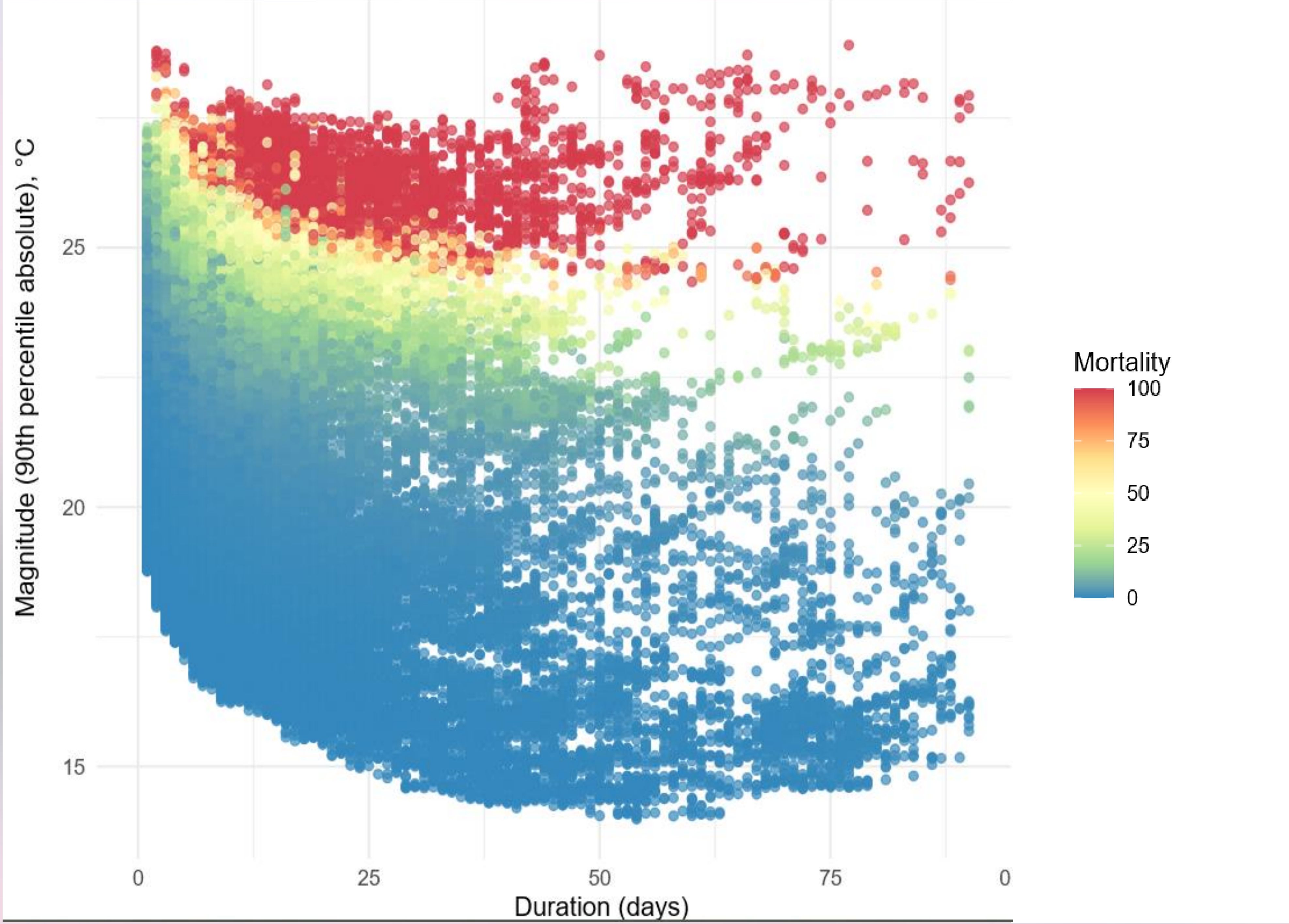
References

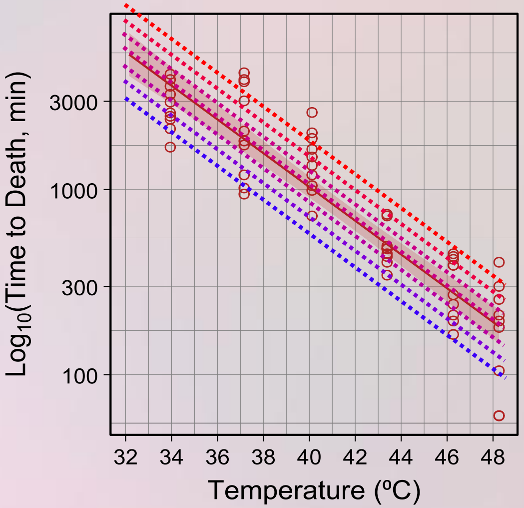
Slides


Environment
Biology
Ecologically
relevant
temperature
Extremity
from an
organismal
perspective
Time-integrated
physiological
performance


Magnitude (90th percentile absolute), °C
25
20
15
0
25
50
75
0
Duration (days)
Mortality
100
75
50
25
0

A BIOLOGICAL APPROACH:
THERMAL TOLERANCE LANDSCAPES

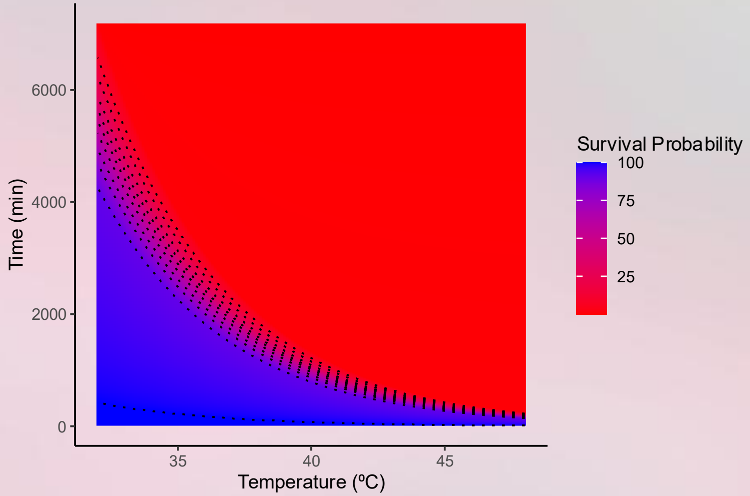
- Important to consider exposure **duration** when measuring thermal tolerance - temperature as a dose.
- Time to death follows a log-linear pattern across taxa (Rezende et al. 2014).
- Creates a **continuous landscape** of survival probability across *acute* and *chronic* timescales.



$$log_{10}time = \frac{CT_{max} - Temp}{z}$$

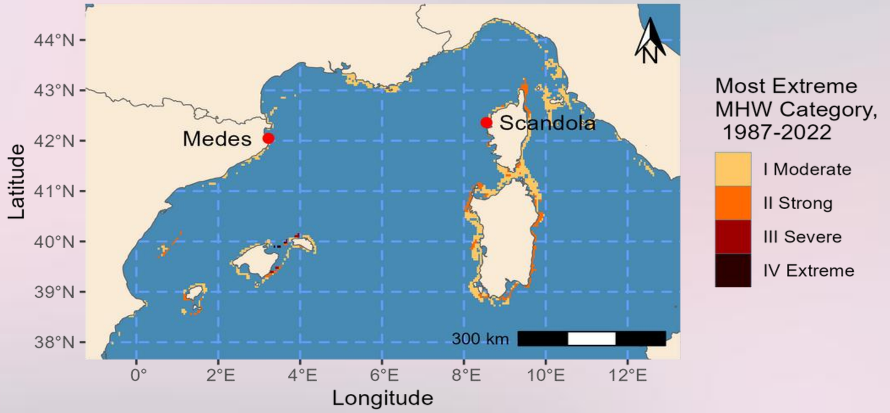
CT_{max} Intercept, when log10 (1 minute) ~ 0

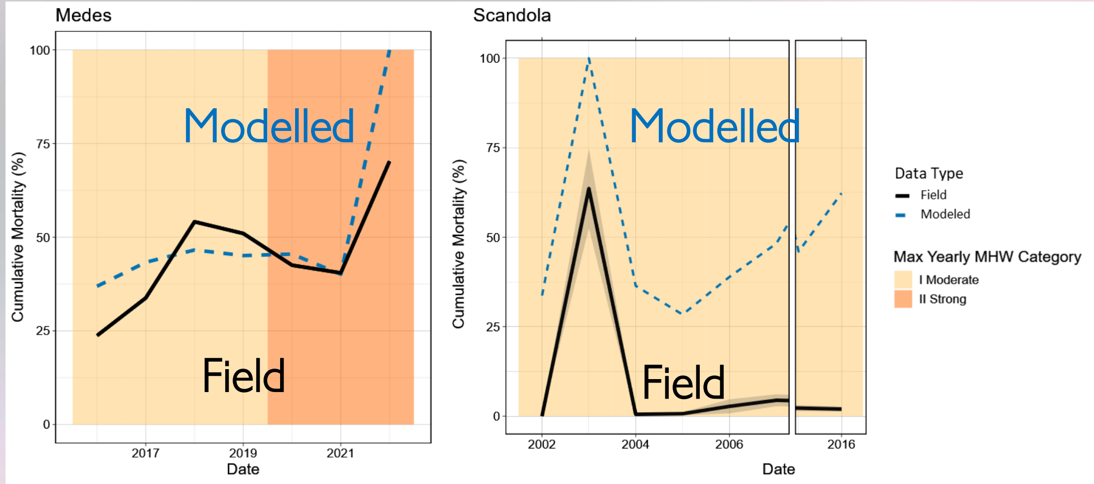
z Negative inverse of slope, scaling factor of tolerance



Rezende et al. 2020 *Science*

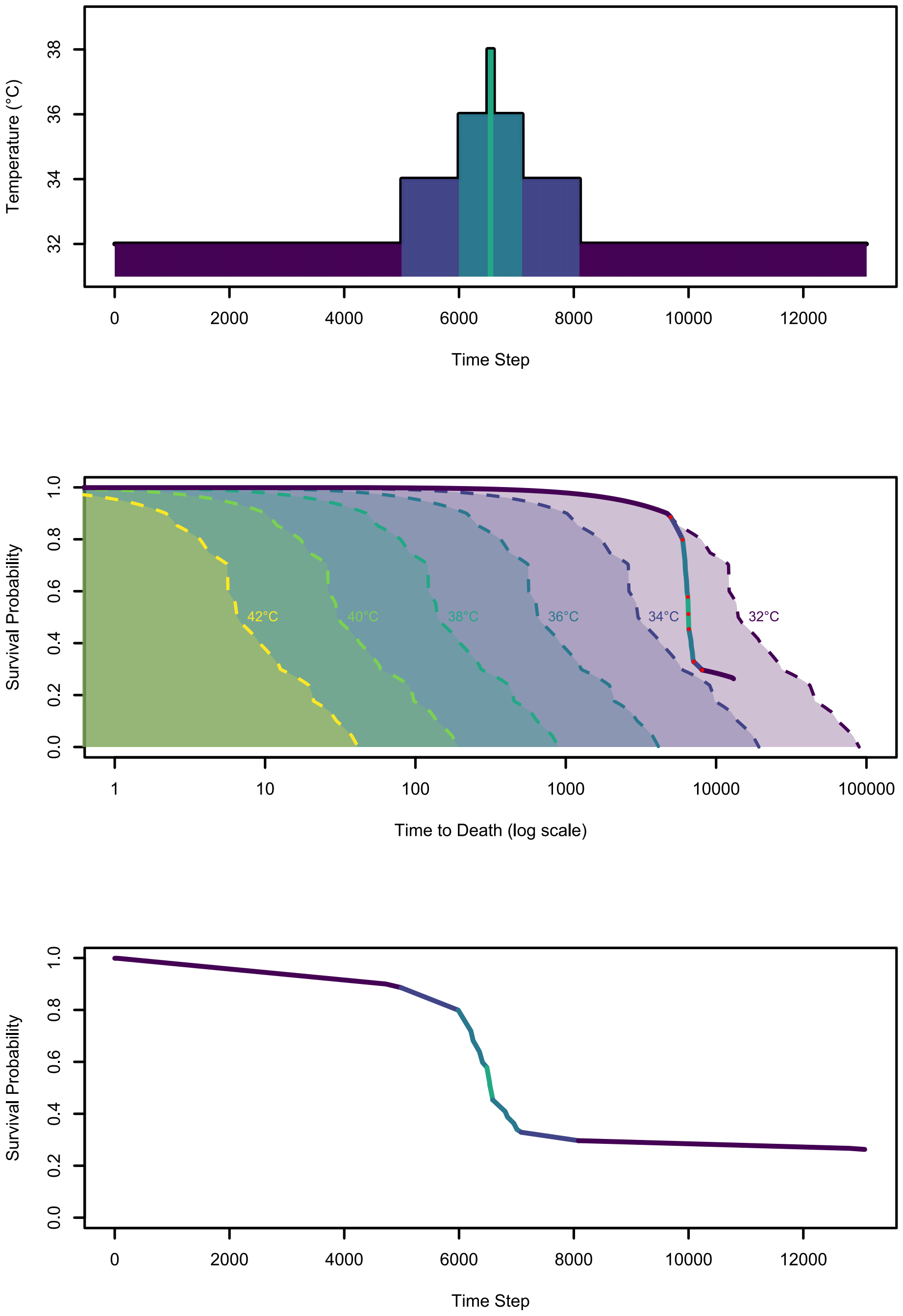
PREDICTIVE ACCURACY VARIES BETWEEN SITES





- Field mortality observed at the Medes Islands, ES, and Scandola MPA, FR and compared against modeled mortality
- Good fit of modeled and observed survival at Medes. Local adaptation may be responsible for overprediction at Scandola.
- Demonstrates inaccuracies in a climatological approach to MHW impacts

Rovira et al. 2024, Gómez-Gras et al. 2021



$$\Delta S_{T(\tau)} \approx \left.\frac{dS(\tau)}{d\tau}\right|_{T(\tau)} \Delta\tau$$